

T3 matched BSM EFT terms by field content

EFT order: 5; loop order: 1. Columns are determined only from matched BSM EFT terms. Empty cells denote no matched term with that field combination.

Field combinations 1 of 2

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$B_{\mu\nu}$	H	$W_{\mu\nu}$	$H \ell$
T3-d3-d5-F4	0	3	0	5	1	4	$\frac{1}{2}$	$ \begin{aligned} & -\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) B_{\mu_1 \mu_2}^2 \\ & -\frac{5}{12} \hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_{S_2}^2} \right) B_{\mu_1 \mu_2}^2 \end{aligned} $	$ \begin{aligned} & +C_{H2} H_{a_1} H_{a_1}^\dagger \\ & -H_{a_1} \mu^2 H_{a_1}^\dagger \\ & +\frac{3\text{lambda}dH1\text{Inv}2^2 h H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon} \\ & +\frac{15\text{lambda}dH2\text{Inv}1^2 h H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{14\epsilon} \\ & +\frac{95\text{lambda}dH2\text{Inv}2^2 h H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{42\epsilon} \\ & +\frac{10\sqrt{\frac{2}{3}}\text{lambda}dH2\text{Inv}1\text{lambda}dH2\text{Inv}2h H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{7\epsilon} \\ & +\frac{3}{2}\text{lambda}dH1\text{Inv}2^2 h \log \left(\frac{\mu_{\text{bar}}^2}{M_{S_1}^2} \right) H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \\ & +\frac{15}{14}\text{lambda}dH2\text{Inv}1^2 h \log \left(\frac{\mu_{\text{bar}}^2}{M_{S_2}^2} \right) H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \\ & +\frac{95}{42}\text{lambda}dH2\text{Inv}2^2 h \log \left(\frac{\mu_{\text{bar}}^2}{M_{S_2}^2} \right) H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \\ & +\frac{10}{7}\sqrt{\frac{2}{3}}\text{lambda}dH2\text{Inv}1\text{lambda}dH2\text{Inv}2h \log \left(\frac{\mu_{\text{bar}}^2}{M_{S_2}^2} \right) H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \\ & +\frac{2\sqrt{\frac{2}{3}}\hbar\lambda_{T3} H_{a_1} H_{a_2} \lambda_{T3} H_{a_1}^\dagger H_{a_2}^\dagger}{\epsilon} \\ & +\frac{2\sqrt{\frac{2}{3}}\hbar\lambda_{T3} H_{a_1} H_{a_2} M_{S_1}^2 \lambda_{T3} H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S_1}^2 - M_{S_2}^2} \\ & +\frac{2\sqrt{\frac{2}{3}}\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_{S_1}^2} \right) \lambda_{T3} H_{a_1} H_{a_2} M_{S_1}^2 \lambda_{T3} H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S_1}^2 - M_{S_2}^2} \\ & -\frac{2\sqrt{\frac{2}{3}}\hbar\lambda_{T3} H_{a_1} H_{a_2} M_{S_1}^2 \lambda_{T3} H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S_1}^2 - M_{S_2}^2} \\ & -\frac{2\sqrt{\frac{2}{3}}\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_{S_2}^2} \right) \lambda_{T3} H_{a_1} H_{a_2} M_{S_2}^2 \lambda_{T3} H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S_1}^2 - M_{S_2}^2} \end{aligned} $	$ \begin{aligned} & -\frac{6\hbar \left(W_{\mu_1 \mu_2}^{A_1} \right)^2}{\epsilon} \\ & -5\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \left(W_{\mu_1 \mu_2}^{A_1} \right)^2 \\ & -\frac{1}{6} \hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_{S_1}^2} \right) \left(W_{\mu_1 \mu_2}^{A_1} \right)^2 \\ & -\frac{5}{6} \hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_{S_2}^2} \right) \left(W_{\mu_1 \mu_2}^{A_1} \right)^2 \end{aligned} $	$ \begin{aligned} & -\frac{2\sqrt{\frac{2}{3}}\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{W_{T1,T1} W_{T2,T2} T_{T1}^T T_{T2}^T} C P_L \ell_{a_2,T1} H_{a_1} H_{a_2} M_{S_1}^2 \lambda_{T1} \ell_{a_1,T2} \ell_{a_2,T4} M_F^2}{(M_F^2 - M_{S_1}^2)(M_F^2 - M_{S_2}^2)(M_{S_1}^2 - M_{S_2}^2)} \\ & +\frac{2\sqrt{\frac{2}{3}}\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{W_{T1,T1} W_{T2,T2} T_{T1}^T T_{T2}^T} C P_L \ell_{a_2,T1} H_{a_1} H_{a_2} M_{S_1}^2 \lambda_{T1} \ell_{a_1,T2} \ell_{a_2,T4} M_F^2}{(M_F^2 - M_{S_1}^2)(M_F^2 - M_{S_2}^2)(M_{S_1}^2 - M_{S_2}^2)} \\ & -\frac{2\sqrt{\frac{2}{3}}\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{W_{T1} \ell_{a_2,T1} C P_R \ell_{a_2,T2}^T} M_{S_1}^2 W_{T1,T1} W_{T2,T2} T_{T1}^T T_{T2}^T \ell_{a_1,T4} \ell_{a_2,T4} H_{a_1}^\dagger H_{a_2}^\dagger M_F^2}{(M_F^2 - M_{S_1}^2)(M_F^2 - M_{S_2}^2)(M_{S_1}^2 - M_{S_2}^2)} \\ & +\frac{2\sqrt{\frac{2}{3}}\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{W_{T1} \ell_{a_2,T1} C P_R \ell_{a_2,T2}^T} M_{S_1}^2 W_{T1,T1} W_{T2,T2} T_{T1}^T T_{T2}^T \ell_{a_1,T4} \ell_{a_2,T4} H_{a_1}^\dagger H_{a_2}^\dagger M_F^2}{(M_F^2 - M_{S_1}^2)(M_F^2 - M_{S_2}^2)(M_{S_1}^2 - M_{S_2}^2)} \\ & -\frac{2\sqrt{\frac{2}{3}}\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{W_{T1,T1} W_{T2,T2} T_{T1}^T T_{T2}^T} C P_L \ell_{a_2,T1} H_{a_1} H_{a_2} M_{S_1}^2 M_{S_1}^2 \lambda_{T1} \ell_{a_1,T2} \ell_{a_2,T4} M_F}{(M_F^2 - M_{S_1}^2)(M_F^2 - M_{S_2}^2)(M_{S_1}^2 - M_{S_2}^2)} \\ & +\frac{2\sqrt{\frac{2}{3}}\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{W_{T1,T1} W_{T2,T2} T_{T1}^T T_{T2}^T} C P_L \ell_{a_2,T1} H_{a_1} H_{a_2} M_{S_1}^2 M_{S_1}^2 \lambda_{T1} \ell_{a_1,T2} \ell_{a_2,T4} M_F}{(M_F^2 - M_{S_1}^2)(M_F^2 - M_{S_2}^2)(M_{S_1}^2 - M_{S_2}^2)} \\ & -\frac{2\sqrt{\frac{2}{3}}\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{W_{T1} \ell_{a_2,T1} C P_R \ell_{a_2,T2}^T} M_{S_1}^2 M_{S_1}^2 W_{T1,T1} W_{T2,T2} T_{T1}^T T_{T2}^T \ell_{a_1,T4} \ell_{a_2,T4} H_{a_1}^\dagger H_{a_2}^\dagger M_F}{(M_F^2 - M_{S_1}^2)(M_F^2 - M_{S_2}^2)(M_{S_1}^2 - M_{S_2}^2)} \\ & +\frac{2\sqrt{\frac{2}{3}}\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{W_{T1} \ell_{a_2,T1} C P_R \ell_{a_2,T2}^T} M_{S_1}^2 M_{S_1}^2 W_{T1,T1} W_{T2,T2} T_{T1}^T T_{T2}^T \ell_{a_1,T4} \ell_{a_2,T4} H_{a_1}^\dagger H_{a_2}^\dagger M_F}{(M_F^2 - M_{S_1}^2)(M_F^2 - M_{S_2}^2)(M_{S_1}^2 - M_{S_2}^2)} \end{aligned} $

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	H	ℓ	e
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